

ARK-445 — RUN LOG

Remnant Fieldworks Inc. — Derek Hone

Tri-State Authorization Discrimination (ALLOW / HOLD / DENY) on ibm_marrakesh (Heron r2).

Protocol: Proof Before Power. Prediction Before Measurement. No Rescue After Failure.

Each step below writes files + a git commit; commit timestamps prove ordering (Field 27).

Step 1 — LOCK (preregistration + code + MANIFEST)

- **LOCK commit SHA:** `b0b56b08e58db02040f30937f487f1076cac4de9`
- **Branch:** `execute/ark-445`
- **Locked at (UTC):** 2026-07-17T01:32:55Z (MANIFEST generation timestamp)
- **Locked files (SHA-256 in MANIFEST.txt):** ARK_445_preregistration.md, README.md, ark_445_select_qubits.py, ark_445_circuits.py, ark_445_spam_job.py, ark_445_submit_ibm.py, ark_445_retrieve.py, ark_445_analysis.py
- **Pre-lock validation (no QPU cost):**
 - Ideal AerSimulator logic check: ALLOW arms $P(Q_P=1) \sim 1.000$; DENY arms ~ 0.000 ; HOLD arms ~ 0.497 ($H_{\text{plus}} \sim H_{\text{minus}}$, symmetric); confusion/replay ~ 0.000 (DENY); SPAM idle ~ 0.000 .
 - Real-backend transpile check on ibm_marrakesh (`opt_level=3`, `seed_transpiler=445`): all 9 principal arms + arm10 transpile cleanly; `if_test ( if_else )` blocks preserved; HOLD H decomposes to `rz/sx`; arm9 reset+measure intact.
- **No IBM Quantum hardware job has been submitted as of the LOCK commit.**

Step 2 — Qubit selection

- (pending)

Step 3-4 — SPAM kill-gate

- (pending)

Step 6-7 — Principal job

- (pending)